

- Q.29 Explain the measurement of pH of liquid. (CO5)
- Q.30 Write short note on viscosity and vibration. (CO5)
- Q.31 Explain the construction and working of LED display. (CO4)
- Q.32 Describe, how temperature can be measured using a resistive thermometer? (CO3)
- Q.33 Compare and contrast different methods of measuring pressure. (CO3)
- Q.34 Explain the concept of torque measurement and its significance in engineering. (CO4)
- Q.35 Discuss the integration of sensors and data acquisition systems in modern instrumentation. (CO5)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Discuss in detail the various methods for measuring displacement, including their principles and applications. (CO2)
- Q.37 Explain the principles and applications of different type's pressure measuring devices. (CO3)
- Q.38 Write short note on any two:
- a) Load cell (CO4)
 - b) Ultrasonic flow meter (CO5)
 - c) Humidity (CO5)

No. of Printed Pages : 4
Roll No.

202434

3rd Semester/ Mechtronics Subject : Electronic Instrumentation

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following transducers converts mechanical displacement into an electrical signal? (CO1)
- a) Thermocouple b) Strain gauge
 - c) Photoresistor d) Capacitor
- Q.2 The unit of force measurement in the SI system is: (CO1)
- a) Joule b) Newton
 - c) Pascal d) Watt
- Q.3 Which type of pressure measurement device uses a diaphragm? (CO4)
- a) Bourdon tube b) Thermocouple
 - c) Piezoelectric sensor d) Capacitive sensor
- Q.4 A flow measurement device that measures the velocity of fluid is called? (CO3)
- a) Manometer b) Orifice plate
 - c) Pitot tube d) Venturi meter

- Q.5 _____ is primarily used for measuring temperature? (CO3)
- a) Strain gauge b) Thermistor
c) LVDT d) Load Cell
- Q.6 Data acquisition system is used to: (CO6)
- a) Generate electrical signals
b) Analyze data manually
c) Collect and process data
d) Control output devices
- Q.7 Which of the following represent active transducer? (CO2)
- a) Strain gauge b) Thermistor
c) LVDT d) Thermocouple
- Q.8 What is the principle of operation of LVDT? (CO3)
- a) Mutual inductance b) Self inductance
c) Permanence d) Reluctance
- Q.9 An accelerometer is used to measure _____. (CO3)
- a) Pressure b) Force
c) Displacement d) Acceleration
- Q.10 The primary purpose of a single conditioning circuit is: (CO4)
- a) Amplify signals b) Generate signals
c) Store signals d) Visualize signals

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Define measurand. (CO1)

(2)

202434

- Q.12 Expand LCD. (CO1)
- Q.13 A _____ is used to convert a physical quantity into an electrical signal. (CO2)
- Q.14 The _____ is a common device for measuring flow in pipelines. (CO2)
- Q.15 A _____ sensor is used to measure temperature variations. (CO3)
- Q.16 The principle of operation of a load cell is based on _____. (CO4)
- Q.17 Define Gauge Factor. (CO3)
- Q.18 Pyrometer is used to measure _____. (CO4)
- Q.19 Define humidity. (CO5)
- Q.20 _____ is used to measure speed. (CO3)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define measurement. Also mention its importance. (CO1)
- Q.22 Explain piezoelectric transducer with its construction. (CO3)
- Q.23 Explain the construction of capacitive transducer. (Co2)
- Q.24 What is DAS? Explain its main components? (CO6)
- Q.25 Explain the working of resistive strain gauge. (CO2)
- Q.26 Describe the operation of dynamometer and its applications in measurement. (CO4)
- Q.27 Explain the basic principle of flow measurement. (CO4)
- Q.28 Explain the operation of bimetallic thermometer. (CO3)

(3)

202434